CLINICAL · NECK, THYROID & SALIVARY

# Head and neck cancer: biology, staging and principles

Head and neck squamous carcinoma is several diseases that share a microscope slide. One is tobacco- and alcohol-driven cancer of the mouth, larynx and hypopharynx in older smokers. Another is HPV-driven cancer of the tonsil and tongue base in younger non-smokers, which carries a far better prognosis. A third is EBV-driven nasopharyngeal carcinoma. The principles here apply across sites: biology, staging by the eighth edition of the AJCC system, the work-up, the multidisciplinary decision, and life after treatment.

**AT A GLANCE**

- Tobacco and alcohol act synergistically.
- HPV: E6 degrades p53, E7 inactivates Rb, p16 rises.
- AJCC 8 added depth of invasion, extranodal extension and p16 staging.
- Treat the N0 neck when occult risk exceeds about 15 to 20%.

## 1 CHAPTER ONE Who and why

Epidemiology and causes.

### 1.1 Burden

Head and neck cancer is about the seventh or eighth commonest cancer worldwide. In India it is among the leading cancers in men, mostly *oral cavity* cancer linked to smokeless tobacco and betel quid. More than 90% of tumours of the mucosa of the upper aerodigestive tract are squamous cell carcinomas, and in many low-resource settings more than 70% of patients present at stage III or IV.

### 1.2 Risk factors

**Tobacco** is the principal cause, whether smoked as cigarettes, bidis or hookah, or used in smokeless forms: chewing tobacco, gutka, *betel quid with areca nut* and snuff. *Reverse smoking*, with the lit end inside the mouth, causes cancer of the palate.

**Alcohol** acts *synergistically* with tobacco, so that the combined risk is multiplicative rather than additive.

**HPV-16** causes oropharyngeal cancer, especially of the tonsil and tongue base; it is linked to oral sex and the number of partners, and its incidence is rising in high-income countries. **Epstein–Barr virus** causes undifferentiated nasopharyngeal carcinoma, together with salted fish and preserved foods and genetic susceptibility in southern China.

Other factors include poor oral hygiene and chronic dental trauma, submucous fibrosis from areca nut, Plummer–Vinson syndrome for postcricoid cancer, immunosuppression, radiation, occupational exposure to wood dust, leather and nickel for sinonasal adenocarcinoma and squamous carcinoma, Fanconi anaemia and dyskeratosis congenita, and diets poor in fruit and vegetables.

## 1.3 Field cancerisation and second primaries

Carcinogens bathe the whole mucosa, so genetically altered fields surround a cancer and give rise to new ones. Second primary tumours develop at about 3 to 5% a year, in the head and neck, lung and oesophagus, and they are a leading cause of death after successful treatment of an early cancer. This is why every patient has a panendoscopy at diagnosis, is helped to stop smoking, and is followed for life.

## 1.4 Premalignant lesions

Leukoplakia and erythroplakia, of which erythroplakia carries the highest risk, submucous fibrosis, erosive lichen planus and laryngeal dysplasia are the recognised premalignant states. The grade of dysplasia, from mild through moderate and severe to carcinoma in situ, predicts the risk of invasion.

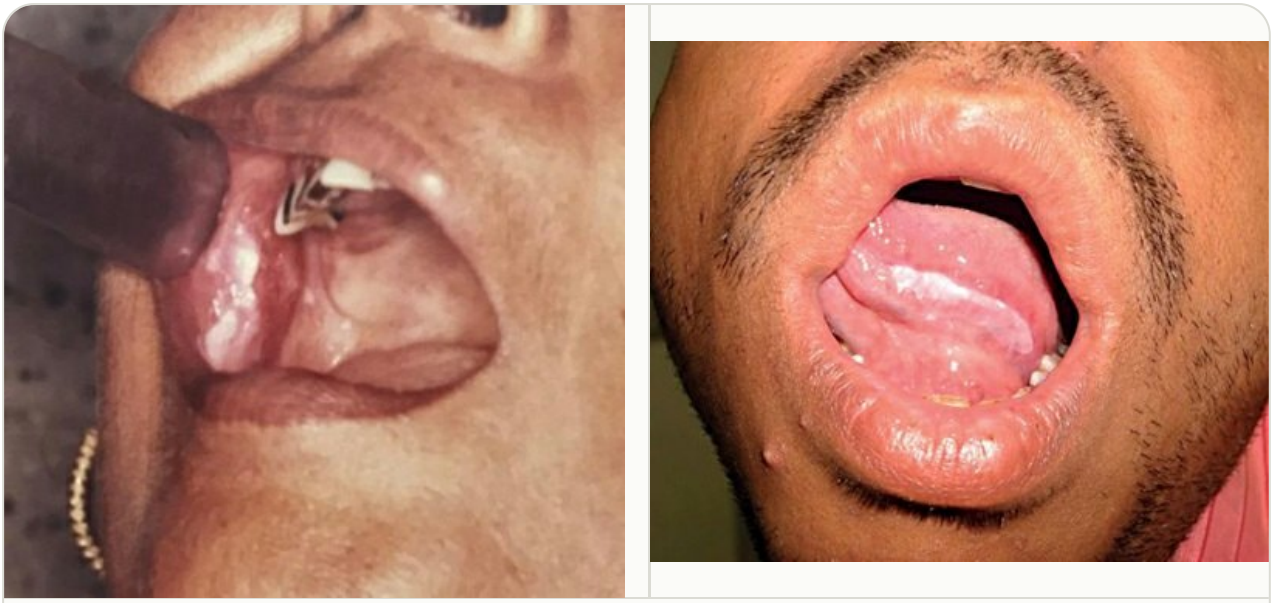


**FIGURE 1** Leukoplakia of the cheek and of the lateral tongue.
Dhingra, figs 44.7 and 44.8

# 2 CHAPTER TWO Tumour biology

Two diseases.

## 2.1 HPV-negative carcinoma

HPV-negative cancer is driven by tobacco carcinogens that cause accumulated mutations. Tumour suppressors are inactivated: *TP53* is mutated in about 80%, *CDKN2A* (p16) is lost by mutation, deletion of 9p21 or methylation, and *FAT1*, *NOTCH1* and *CASP8* are often affected. Oncogenes are activated: *cyclin D1* by 11q13 amplification, *PIK3CA* at 3q26, *EGFR*, which is overexpressed in about 90% and amplified in some, and *FGFR1*, *MYC* and *HRAS*. Califano's progression model describes the sequence: loss of 9p21 (p16), then 3p, then 17p (p53), then further changes, as normal mucosa moves through dysplasia to invasive cancer.

## 2.2 HPV-positive carcinoma

In HPV-positive cancer the viral oncoprotein *E6* degrades *p53* and *E7* inactivates *Rb*. With Rb gone, *p16 is strongly over-expressed*, which is why *p16 immunohistochemistry* serves as the surrogate marker for HPV in oropharyngeal cancer. p53 is usually *wild-*

*type*, the mutational burden is lower, *PIK3CA* activation is frequent, and changes in *TRAF3* and *CYLD* are linked to immune escape.

Clinically the patients are younger and often non-smokers, with small primaries in the tonsil or tongue base and early, often *cystic*, nodes. The tumours respond much better to radiotherapy and chemotherapy, with about 80% three-year survival against 55% for HPV-negative tumours in RTOG 0129, although smoking more than 10 pack-years worsens the outlook.

## 2.3 Prognostic factors

The tumour factors that matter are stage, especially nodal disease and *extranodal extension*; *depth of invasion* in oral cancer; perineural and lymphovascular invasion; the margins; and HPV status. Comorbidity, performance status and continued smoking are the patient factors, and total treatment time and gaps in radiotherapy the treatment factors.

# 3 CHAPTER THREE Staging

AJCC eighth edition.

## 3.1 Principles

The **T** stage describes the primary, mostly by size, with important exceptions: vocal fold mobility and subsite spread in the larynx, *depth of invasion* in the oral cavity, and extension in the nasopharynx and sinuses. The **N** stage describes regional nodes by size, number and side and now by *extranodal extension* (ENE), and the **M** stage distant metastases. Clinical staging (cTNM) comes from examination and imaging, pathological staging (pTNM) from the resected specimen.

T4 is subdivided into *T4a*, moderately advanced and resectable, and *T4b*, very advanced and usually unresectable, involving for example the masticator space, pterygoid plates, skull base, prevertebral fascia or mediastinum, or encasing the carotid.

### T STAGE: ORAL CAVITY (INCLUDING LIP)

| T | DEFINITION |
|---|---|
| T1 | 2 cm or less, depth of invasion (DOI) 5 mm or less |
| T2 | 2 cm or less with DOI over 5 mm, or over 2 to 4 cm with DOI 10 mm or less |
| T3 | Over 4 cm, or DOI over 10 mm |
| T4a | Invades through cortical bone of mandible or maxilla, the maxillary sinus or facial skin (lip: through bone, inferior alveolar nerve, floor of mouth or skin) |
| T4b | Masticator space, pterygoid plates, skull base, or encases the internal carotid |

### T STAGE: LARYNX (GLOTTIS)

| T | DEFINITION |
|---|---|
| T1a / T1b | One vocal fold / both vocal folds, normal mobility |
| T2 | Extends to supraglottis or subglottis, or impaired mobility |
| T3 | Fixed vocal fold, paraglottic space or inner cortex of thyroid cartilage |
| T4a | Through the outer cortex of the thyroid cartilage or beyond the larynx (trachea, strap muscles, thyroid, oesophagus, deep tongue muscles) |
| T4b | Prevertebral space, encases carotid, or mediastinum |

The other sites follow the same logic. In the **supraglottis**, T1 involves one subsite, T2 more than one subsite or the glottis or adjacent mucosa, and T3 a fixed fold or the postcricoid area, pre-epiglottic or paraglottic space or minor cartilage erosion, with T4 as for the glottis. In the **hypopharynx**, T1 is 2 cm or less in one subsite, T2 over 2 to 4 cm or more than one subsite, T3 over 4 cm or with a fixed hemilarynx or oesophageal mucosal spread, and T4a invades cartilage, hyoid, thyroid, oesophageal muscle or the central compartment. In the **oropharynx**, for both HPV types, T1 is 2 cm or less, T2 is 2 to 4 cm and T3 is over 4 cm or reaches the lingual surface of the epiglottis, and HPV-positive tumours have no T4b. In the **nasopharynx**, T1 is confined to the nasopharynx or extends to the oropharynx or nasal cavity, T2 reaches the parapharyngeal space or adjacent muscles, T3 involves the bone of the skull base, cervical vertebrae, pterygoid structures or sinuses, and

T4 extends intracranially or to the cranial nerves, hypopharynx, orbit, parotid or beyond the lateral pterygoid.

## 3.2 Nodal staging

N STAGE: CLINICAL (ORAL CAVITY, LARYNX, HYPOPHARYNX, HPV-NEGATIVE OROPHARYNX)

| N | DEFINITION |
|---|---|
| N1 | Single ipsilateral node 3 cm or less, ENE negative |
| N2a | Single ipsilateral node over 3 to 6 cm, ENE negative |
| N2b | Multiple ipsilateral nodes, none over 6 cm, ENE negative |
| N2c | Bilateral or contralateral nodes, none over 6 cm, ENE negative |
| N3a | Any node over 6 cm, ENE negative |
| N3b | Clinically overt ENE (skin involvement, fixation, cranial nerve or muscle invasion) |

Pathologically, a single ipsilateral node of 3 cm or less *with* ENE becomes pN2a, and larger or multiple nodes with ENE become pN3b. In the **HPV-positive oropharynx**, clinical N1 is ipsilateral nodes of 6 cm or less, N2 contralateral or bilateral nodes of 6 cm or less, and N3 any node over 6 cm, while pathologically *pN1 is one to four nodes and pN2 five or more*. In the **nasopharynx**, N1 is unilateral cervical, or unilateral or bilateral retropharyngeal nodes, of 6 cm or less above the cricoid; N2 is bilateral cervical nodes of 6 cm or less above the cricoid; and N3 is any node over 6 cm or below the caudal border of the cricoid.

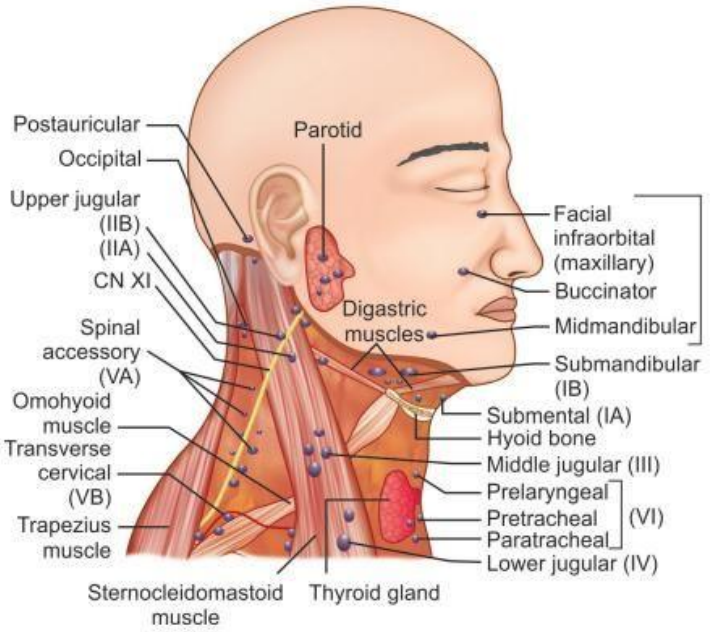



**Fig. 6:** Lymph nodes of head and neck. Neck nodes regions (levels) are mentioned in parenthesis

**FIGURE 2** Lymph node levels of the neck.

Anatomy atlas · ENT Steps p. 285

## 3.3 Stage groups

### STAGE GROUPS (ORAL CAVITY, LARYNX, HYPOPHARYNX, HPV-NEGATIVE OROPHARYNX, SINUSES, SALIVARY)

| STAGE | TNM |
|---|---|
| I | T1 N0 M0 |
| II | T2 N0 M0 |
| III | T3 N0, or T1 to T3 N1, M0 |
| IVA | T4a N0 to N1, or T1 to T4a N2, M0 |
| IVB | Any T N3, or T4b any N, M0 |
| IVC | Any M1 |

HPV-positive oropharyngeal cancer has its own clinical groups: stage I is T0 to T2 with N0 to N1, stage II is T0 to T2 with N2 or T3 with N0 to N2, stage III is T4 or N3, and stage IV is M1. Many patients are downstaged compared with the seventh edition, which reflects their better prognosis.

**THE EIGHTH EDITION CHANGES**

The 2017 AJCC eighth edition introduced:

- *Depth of invasion* for oral cancer.
- *Extranodal extension* in nodal staging.
- A separate staging system for *p16-positive oropharyngeal cancer*.
- Separate staging for unknown primaries by EBV and HPV status.

Older texts and exam questions based on the seventh edition will give different stages for the same tumour.

# 4 CHAPTER FOUR Work-up

Before the decision.

## 4.1 Assessment

The history covers symptoms, risk factors, comorbidity, performance status (ECOG or Karnofsky), nutrition and social support. The examination covers the whole head and neck, with palpation of the tongue base and floor of mouth, flexible nasendoscopy, the cranial nerves, and the neck by level. Tissue comes from biopsy of the primary and FNA of nodes, with p16/HPV testing for the oropharynx and unknown primaries and EBV for the nasopharynx.

## 4.2 Imaging

CT and MRI of the primary and neck are complementary: MRI is better for the tongue, nasopharynx, skull base and perineural spread, CT for bone and the larynx. Distant staging uses CT of the chest for advanced disease and *PET-CT* for stage III and IV disease, unknown primaries and recurrence, and the neck is assessed with ultrasound and FNA.

## 4.3 Panendoscopy

Examination under anaesthesia, with laryngoscopy, pharyngoscopy, oesophagoscopy and sometimes bronchoscopy, maps the tumour's extent into the tongue base, vallecula, postcricoid area and subglottis, takes biopsies, and searches for synchronous second primaries. For unknown primaries it includes tonsillectomy and tongue base mucosectomy.

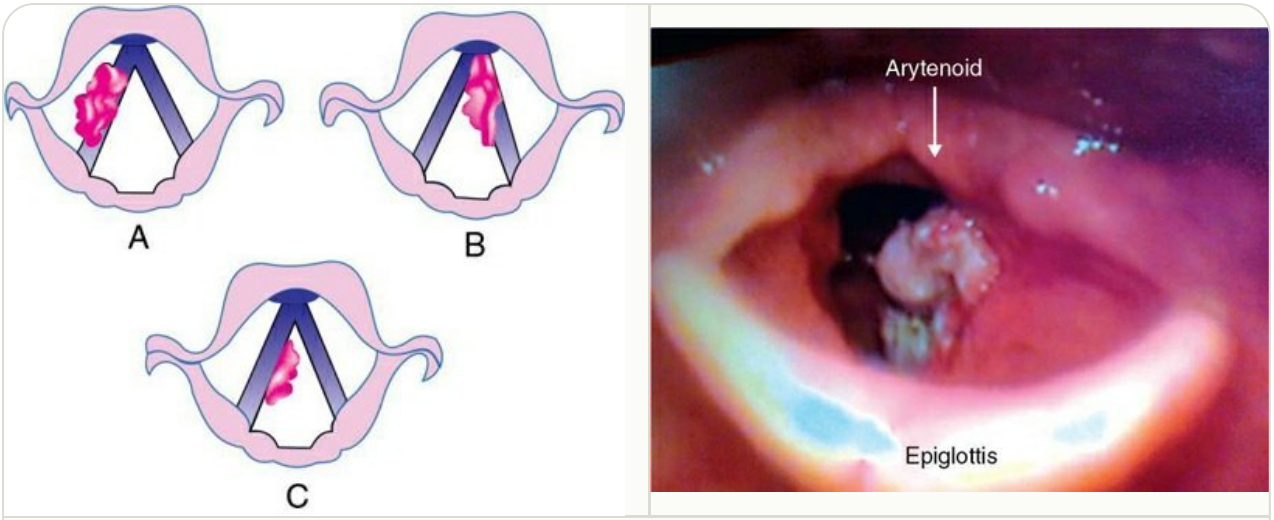



**FIGURE 3** Laryngeal cancer by subsite: supraglottic, glottic and subglottic.
Dhingra, figs 62.4 and 62.5

## 4.4 Before treatment

Before treatment begins, teeth at risk are extracted ahead of radiotherapy, a dietitian reviews nutrition and a prophylactic gastrostomy is considered, and a speech and swallowing therapist makes a baseline assessment and starts *prehabilitation*. Smoking and alcohol cessation are supported, the anaesthetist plans the airway (some patients need a tracheostomy under local anaesthesia first), and the clinical nurse specialist begins psychosocial support.

**OPERATIVE GUIDES IN THEATRE** 2 operations

Management of the unknown primary, and panendoscopy.

**Management of the Unknown Primary Cancer**
Unknown primary

**Esophagoscopy and Hypopharyngoscopy**
Panendoscopy

# 5 CHAPTER FIVE Treatment

The multidisciplinary decision.

## 5.1 The team

Every patient is discussed at a *multidisciplinary team (tumour board)* of head and neck and reconstructive surgeons, clinical and medical oncologists, radiologist, pathologist, restorative dentist, speech and language therapist, dietitian, specialist nurse and palliative care. The patient's own priorities are central: cure, voice, swallowing, appearance and time.

## 5.2 General principles

**Early disease (stages I and II)** is treated with a single modality, surgery *or* radiotherapy, which give similar cure rates. The choice is driven by function, access, morbidity, the patient's wishes and the wish to keep options open for second primaries: transoral laser surgery or radiotherapy for a T1 glottic cancer, for example, or surgery for a small oral tongue cancer.

**Advanced disease (stages III and IV)** needs combined treatment. *Surgery followed by radiotherapy*, or chemoradiotherapy for positive margins or extranodal extension, is preferred for oral cavity cancer and for cancers destroying cartilage or bone, such as T4a larynx and mandibular invasion. *Primary concurrent chemoradiotherapy* with surgical salvage is the organ-preservation approach for the larynx, hypopharynx and oropharynx, and the standard for the nasopharynx.

**Unresectable T4b disease** is treated with chemoradiotherapy or palliation. **Recurrent or metastatic disease** is treated by salvage surgery or re-irradiation for local and regional recurrence, systemic therapy with pembrolizumab with or without platinum and 5-FU or cetuximab-based regimens, and palliative care.

### 5.3 The neck

A **clinically node-negative neck** is treated electively, by selective neck dissection or elective radiotherapy, when the risk of occult nodes exceeds about 15 to 20%. In the oral cavity this means a depth of invasion over 3 to 4 mm, and the D'Cruz trial showed a survival advantage for elective neck dissection in early oral cancer; sentinel node biopsy is an alternative. A **clinically positive neck** is treated by therapeutic neck dissection, modified radical or comprehensive, or by chemoradiotherapy with a PET-CT at 12 weeks deciding whether dissection is needed. Tumours of the tongue base, soft palate, supraglottis and floor of mouth crossing the midline drain to *both* sides, and both necks are treated.

### 5.4 Surgery

The aim of surgery is complete resection with clear margins, generally *5 mm or more* on histology, a margin under 5 mm being close; frozen sections help. The approaches range from transoral laser microsurgery and transoral robotic surgery to the lip-split mandibulotomy, lateral pharyngotomy, total laryngectomy and pharyngolaryngectomy. Reconstruction usually takes place at the same operation, often with free flaps, and a tracheostomy protects the airway after major oral and pharyngeal resections.

## 6 CHAPTER SIX After treatment

Survival and survivorship.

### 6.1 Follow-up

Most recurrences occur within the first *two years*, so patients are seen every 1 to 3 months in the first year, every 2 to 4 months in the second, every 3 to 6 months in years three to five, and yearly thereafter. Each visit includes examination and nasendoscopy, with a PET-CT about 12 weeks after radiotherapy and further imaging as indicated. *TSH* is checked every 6 to 12 months after neck radiotherapy,

the teeth are reviewed, smoking status is recorded, and the whole upper aerodigestive tract and chest are watched for second primaries.

## 6.2 Rehabilitation and quality of life

Swallowing is rehabilitated with exercises, dilatation of strictures, and a gastrostomy for those who cannot eat safely. After laryngectomy, voice is restored with a *tracheo-oesophageal valve*, an electrolarynx or oesophageal speech. Shoulder dysfunction after accessory nerve injury needs physiotherapy and lymphoedema manual drainage, and trismus, xerostomia, hearing loss and osteoradionecrosis are the other late effects to manage.

Depression, fear of recurrence, alcohol relapse and suicide are all common in this group, so patients are screened and referred. Palliative care controls symptoms, manages a fungating tumour, and plans for *carotid blow-out*, airway obstruction and bleeding, with a crisis plan of dark towels and sedation.

# 7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

## 7.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Risk factors:** tobacco (including smokeless) and alcohol synergy; HPV; EBV.
- **Field cancerisation** and the second primary rate.
- **HPV-positive versus HPV-negative cancer:** E6 degrades p53, E7 inactivates Rb so p16 is over-expressed; prognosis; a separate AJCC staging system.
- **The eighth edition changes:** depth of invasion, ENE, p16 staging.
- **Oral cavity and glottic T staging;** the clinical N stages; the stage groups.
- **The work-up:** imaging, PET-CT, panendoscopy, the dental and nutrition review.
- **Early versus advanced disease:** single versus combined modality; when to treat the N0 neck; margins.

- **Follow-up schedule** and late effects.

## 7.2 Common pitfalls

COMMON PITFALLS

- Staging an oropharyngeal cancer without p16.
- Using seventh-edition definitions in an eighth-edition world.
- Forgetting the contralateral neck in midline tumours.
- Neglecting dental assessment and smoking cessation before treatment.
- Discharging patients without thyroid and second primary surveillance.

Written for this app from Oxford Textbook of Otolaryngology ch. 68, 69, Oxford Specialist Handbook ch. 4, 12, Oxford Handbook of ENT ch. 14, K. J. Lee ch. 41, 48 and Dhingra ch. 73. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.